LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various locking devices, Fasteners and fork joints
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various locking devices, Fasteners and fork joints
5. Specific Objectives	9.1.2.1 Trainee will be able to explain the locking and swaging etc.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of locking devices on fasteners and assembly of fork joints; Safety: Wear eye protection and gloves; Instructor supervision: Monitor tool handling and torque application.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a locking device?
(b) Name two common types of aircraft fasteners.
(c) Why is vibration resistance critical in engine health monitoring?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of locking devices • Mechanism of locking washers • Locking nuts and prevailing torque • Safety wire installation • Swaging basics	L & Ds	PPT, IB	15 min
(b)	• Fork joint construction • Pin and clevis assembly • Locking methods for pins • Inspection criteria • Common failure modes	L & Ds	PPT, PDF	20 min
(c)	• Selection of locking devices • Installation procedures • Torque specifications • Documentation requirements • Maintenance checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a safety wire prevents fastener loosening.
(b) Describe the difference between a split lock washer and a tooth lock washer.
(c) What are the key inspection points for a fork joint assembly?

12. Home Assignment and Follow Up	Prepare a brief note on the tools required for the swaging process discussed in the next lesson plan (LP 07) and list three safety precautions specific to that procedure.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Swaging, plumbing and flaring process and tools used	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min